

GADAG, India

WMO: 432010

Lat: 15.4175N

Lon: 75.6444E

Elev: 2133

StdP: 13.60

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	60.1	61.9	31.0	27.3	83.7	35.4	32.8	83.6	16.1	77.8	14.0	78.4	2.2	140	0.284

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	23.0	99.7	68.7	98.1	68.8	96.4	68.9	77.7	86.4	76.5	84.8	75.5	83.6	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	145.0	81.5	74.4	139.3	80.4	73.4	134.7	79.5	43.0	87.5	41.8	84.6	40.8	83.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.0	13.8	12.0	DB	55.3	101.9	1.8	1.6	54.0	103.0	53.0	104.0	51.9	104.9	50.6	106.1	
			WB	47.3	80.0	1.6	2.0	46.1	81.4	45.2	82.5	44.3	83.6	43.1	85.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	74.4	78.8	83.6	85.7	84.8	79.2	76.9	76.6	77.3	77.8	75.7	73.5
	DBStd	4.54	2.46	2.86	2.76	2.31	2.66	2.80	1.96	1.89	2.06	2.14	2.36	2.55
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10465	757	806	1041	1072	1078	875	834	823	818	862	772	727
	CDD65	4990	292	386	576	622	613	425	369	358	368	397	322	262
	CDH74	44841	2461	4054	7122	7818	7055	3039	1898	1794	2215	2980	2406	1999
CDH80	19456	788	1917	3926	4460	3813	1019	396	364	597	985	685	506	
Wind	WSAvg	5.0	3.5	3.5	3.6	4.2	5.7	7.7	8.1	6.6	5.0	3.6	4.1	3.9
Precipitation	PrecAvg	26.3	0.0	0.1	0.6	1.6	3.2	3.4	3.2	3.4	4.3	4.4	1.7	0.3
	PrecMax	41.4	0.4	0.8	8.5	4.7	5.6	10.0	10.5	13.0	12.4	15.7	7.4	3.5
	PrecMin	15.5	0.0	0.0	0.0	0.5	0.4	0.8	0.9	0.9	0.8	0.1	0.1	0.0
	PrecStd	6.4	0.1	0.2	1.6	1.1	1.3	2.1	2.0	2.6	2.7	3.2	2.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	95.5	99.4	102.0	101.7	95.1	88.2	88.2	90.3	90.3	88.6	88.7
		MCWB	64.8	65.4	67.7	68.8	67.8	72.0	71.5	71.4	69.2	67.6	67.1	66.6
	2%	DB	87.9	93.5	97.7	100.1	99.7	91.1	85.8	85.7	87.8	88.6	86.9	86.2
		MCWB	63.9	65.1	67.9	68.9	68.9	73.1	71.7	71.8	70.0	67.6	67.2	65.6
	5%	DB	86.1	91.6	96.2	98.6	97.8	88.4	84.1	83.9	85.7	87.1	85.4	84.5
		MCWB	63.2	64.9	67.5	68.7	69.6	73.2	71.9	71.7	70.9	68.0	67.2	64.4
	10%	DB	84.5	89.7	94.6	96.8	95.4	85.7	82.3	82.0	83.6	85.5	84.1	82.9
		MCWB	63.2	64.6	66.5	69.1	70.3	73.2	72.0	71.8	71.3	68.6	66.8	64.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	74.6	79.2	79.6	79.3	77.8	76.1	76.2	76.0	76.2	74.6	73.4
		MCDB	82.2	86.6	91.2	91.0	87.4	84.4	80.9	81.3	81.3	82.7	81.6	81.9
	2%	WB	70.0	72.1	76.1	77.1	77.6	76.2	74.6	74.6	74.6	74.3	72.8	71.2
		MCDB	80.1	84.9	87.0	86.9	86.1	83.6	80.3	80.4	80.9	80.9	79.8	79.8
	5%	WB	68.3	70.0	74.0	75.6	76.4	75.2	73.6	73.6	73.7	73.3	71.7	69.5
		MCDB	79.9	82.1	84.5	85.0	85.1	82.7	79.6	79.9	80.2	79.7	78.9	77.8
	10%	WB	66.6	68.3	72.4	74.4	75.3	74.4	72.8	72.7	72.9	72.5	70.7	68.1
		MCDB	78.8	81.5	83.3	84.1	84.0	81.6	78.6	78.8	79.2	78.7	78.0	77.2
Mean Daily Temperature Range	5% DB	MDBR	21.1	22.1	22.3	23.0	21.8	13.4	10.9	11.3	13.1	15.1	17.4	20.1
		MCDBR	22.7	23.5	23.4	24.6	24.6	17.7	14.0	14.4	16.7	17.9	19.2	21.7
		MCWBR	9.0	9.3	9.4	8.4	7.4	4.9	4.0	4.4	4.7	6.5	7.6	8.7
	5% WB	MCDBR	19.0	21.2	22.0	22.7	21.1	14.4	11.0	11.3	12.4	13.5	14.5	17.2
		MCWBR	9.3	9.6	9.9	8.8	7.7	6.0	4.9	5.5	5.4	6.0	6.7	8.4
Clear-Sky Solar Irradiance	taub		0.412	0.431	0.505	0.566	0.554	0.505	0.510	0.490	0.469	0.513	0.460	0.430
	taud		2.208	2.177	1.986	1.873	1.926	2.114	2.095	2.164	2.230	2.048	2.153	2.185
	Ebn at Noon		278	278	260	244	243	252	252	259	266	251	260	268
	Edh at Noon		44	47	59	66	62	50	52	49	45	53	46	44
All-Sky Solar Radiation	RadAvg		1671	1921	2072	2131	2068	1569	1357	1396	1517	1581	1559	1575
	RadStd		69	67	80	75	95	121	106	77	125	82	100	67

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.25	N/A	+1.10	+0.61	+0.70	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page